De Morgan's first theorem states that the complement of a product is equal to the sum of the complements: $(A \cdot B)' = A' + B'$. The second theorem states that the complement of a sum is equal to the product of the comp $(A + B)' = A' \cdot B'$.

Chapter 3 Number Systems

To convert a binary number to decimal, expand as a sum of powers of two.

Example: 1011012 = 1*32 + 0*16 + 1*8 + 1*4 + 0*2 + 1*1 = 45 decimal.

Chapter 4 Combinational Circuits

A 4-bit ripple carry adder is built from four full adders. The carry of stage i is connected to stage i+1.
Each full adder computes sum = A xor B xor Cin and carry = majority(A, B, Cin).

A 4-variable Karnaugh map has 16 cells. Adjacent 1s are grouped in powers of two to obtain a simplified SOP ex

SRAM uses latches and is faster. DRAM uses capacitors, needs refresh, and is denser. SRAM is used for cache; D